



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 323

Midterm Examination

Date: 17/12/2025

Total Time: 1 Hour 30 Minutes

Course Title: Numerical Analysis

Semester: Fall 2025

Total Marks: 30

Answer any Three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

- | No. | Questions | CLOs | Marks |
|-----|--|------|-------|
| Q1. | (a) Explain the difference between algebraic and transcendental equations with suitable examples. | CLO1 | [4] |
| | (b) Use the Bisection Method to find a real root of the equation.
$f(x) = 3x - \cos(x) - 1$ | CLO2 | [6] |
| | Correct up to two decimal places with an interval [0, 1]. Also, compute the percentage relative error using the exact solution. | | |
| Q2. | (a) Derive the Newton-Raphson iterative formula for solving equations of the form $f(x) = 0$. | CLO2 | [5] |
| | (b) Apply the Newton-Raphson Method to find the root of the equation.
$f(x) = x \cos x - 2x^2 + 3$ | CLO3 | [5] |
| | Correct up to three iterations, starting with the initial guess $x_0 = 1$. | | |
| Q3. | The following data shows the value of $y = \sqrt{\sec^2 x - 1}$ for $0.1 \leq x \leq 0.3$.
Find | CLO3 | [10] |
| | $y(0.10) = 0.1745$ | | |
| | $y(0.15) = 0.2618$ | | |
| | $y(0.20) = 0.3491$ | | |
| | $y(0.25) = 0.4363$ | | |
| | $y(0.30) = 0.5236$ | | |
| | a) Obtain the values of $y(0.12)$ by Newton's Forward Interpolation Method. | | |
| | b) Obtain the values of $y(0.26)$ by Newton's Backward Interpolation Method. | | |
| Q4. | Comment on which formula provides more accurate results and why.
A robotics research lab is programming a three-joint robotic arm. The position of the robot's end-effector depends on solving the following linear system: | CLO3 | [10] |
| | $5x + 2y + 3z = 1$ | | |
| | $2x + 2y = 7$ | | |
| | $8y + 7z = 14$ | | |

Use the Gauss-Jordan Elimination Method to determine the values of x , y , and z so the robotic arm reaches its target position. Discuss whether this method is computationally efficient for large systems.



Q5. (a) Use four segments to calculate the function

CLO3 [10]

$$\int_0^2 (0.2 + 25x + 3x^2 + 2x^4) dx$$

by using the trapezoidal and Simpson's 3/8 rule to evaluate the integral. Compute the relative percentage error between the numerical and analytical solution. Comment on which method is more accurate and explain why.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
Program B.Sc. in Computer Science and Engineering
MGT 329: Industrial and Operational Management
Mid-Term Examination
Fall 2025

24th Batch
14.12.2025

6th Semester
Time: 1 Hour 30 Minutes

Total Marks: 30

Instructions:

- Answer any **THREE SETS** of questions chronologically.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.

No.	Questions/Statement	Marks
1	a) Define operations management. Give suitable examples of operations management in diverse business fields. What are the key components of operations management?	6
	b) Critically analyze the difference between the production of goods and providing services.	4
2	a) What stages are involved in a product life cycle? Explain it with the help of a diagram.	5
	b) What are the key decision areas in operations strategy, and how do these decisions areas affect the overall operations strategy of an organization?	5
3	a) What are the various types of production systems?	2
	b) Calculate the productivity for the following operations: I. Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. II. A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labor, \$1,000 for materials, and \$300 for overhead.	8
4	a) What do you mean by productivity and efficiency?	2
	b) Student tuition at Boehring University is \$150 per semester credit hour. The state supplements school revenue by \$100 per semester credit hour. The average class size for a typical 3-credit course is 50 students. Labor	8

	costs are \$4,000 per class, material costs are \$20 per student per class, and overhead costs are \$25,000 per class.	
--	--	--

- | | | |
|--|--|--|
| | <ol style="list-style-type: none">What is the <i>multifactor</i> productivity ratio for this course process?If instructors work an average of <u>14</u> hours per week for <u>16</u> weeks for each <u>3</u>-credit class of <u>50</u> students, what is the <i>labor</i> productivity ratio? | |
|--|--|--|



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 325 : Computer Peripherals and Interfacing

Batch: 24th

Date: 11/12/2025

Midterm Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

-
- | | |
|---|------------|
| Answer any five questions. | 5 × 6 = 30 |
| ✓ (a) Define computer peripherals. Explain the term with proper example. | CLO1 2 |
| (b) Classify different types of peripherals and describe each type with suitable examples. | CLO1 4 |
| ✓ (a) Describe the importance of peripherals in a computer system. | CLO1 3 |
| (b) Classify the keyboard based on the types of switches used in its keys. Explain each category with appropriate examples. | CLO1 3 |
| 3. (a) List the different groups of keys on a standard keyboard and briefly describe each group. | CLO1 2 |
| (b) Describe the working principle of the monitor that uses electron gun. | CLO1 4 |
| ✓ (a) State the purpose of the following common memory pins:
i) CS ii) WE | CLO5 2 |
| (b) A memory device contains a pin labeled 'WE'. Determine whether this device is RAM or ROM, and justify your answer. | CLO4 2 |
| (c) A 10-bit ROM contains 2,029 address locations. Calculate the required number of address pins and data pins. | CLO4 2 |
| 5. (a) Define the following terms.
i) ROM ii) EPROM iii) Address decoder | CLO5 3 |
| (b) What are the basic types of pins used for interfacing in a memory device? | CLO4 3 |
| ✓ 6. (a) Draw and describe the pin diagram of Intel 2716 EPROM. | CLO4 3 |
| (b) Mention and describe the operating modes of the Intel 2716 EPROM. | CLO4 3 |
| 7. (a) Describe the operation of a NAND-gate decoder with a suitable diagram. Determine the address range based on your diagram. | CLO5 3 |
| (b) Explain how a decoder can be used for address decoding when interfacing multiple ROM chips. Provide an appropriate diagram and specify the address ranges for the ROMs. | CLO5 3 |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE323: System Analysis & Design

Batch: 24th

Date: 09-12-2025

Mid-Term Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

CLOs Marks

1. (a) Who are the typical stack holders of an Information System? Describe them with example. CLO1 3
(b) Summarize different types of users of a system. CLO1 3
 2. (a) Explain entity, associative entity and attributive entity with example. CLO1 3
(b) Explain system builder's view of knowledge of a system. CLO1 3
 3. (a) Define the term **Organizational Boundary**. CLO1 2
(b) List and briefly define the seven phases of the systems development life cycle (SDLC). CLO1 4
 4. (a) List some business questions that can help a system analyst to identify the business objectives of a system. CLO1 3
(b) Explain different types of feasibility analysis needed before the initiation of a system. CLO1 3
 5. (a) What is the purpose of using analogies when estimating project time? CLO1 2
(b) Using the data provided in the tables for costs and benefits over six years, calculate the present value with a discount rate of 11%. CLO3 4
- | Year | 1 | 2 | 3 | 4 | 5 | 6 |
|----------|-------|-------|-------|-------|-------|-------|
| Costs | 35000 | 40000 | 45000 | 48000 | 52000 | 55000 |
| Benefits | 20000 | 28000 | 50000 | 60000 | 75000 | 84000 |
6. (a) If an organization does not have an updated computer hardware inventory, what essential information should a system analyst collect while setting up a new hardware inventory system? CLO2 3
(b) When evaluating vendor support for computer hardware, what major service categories should an organization consider, and what specific support offerings typically fall under each category? Explain briefly. CLO2 3



7. The cost of developing a website over a six-month period as follows:

CLO3

6

At the end of	Estimated Duration	Estimated Cost	Cumulative Cost	Stage Completed	Actual Cost of Stage to Date	Actual Cost of Project to Date
Month 1	1 month	5000\$	5000\$	100%	5000\$	5000\$
Month 2	1 month	4800\$	9800\$	100%	4800\$	9800\$
Month 3	1 month	4400\$	14200\$	100%	4400\$	14200\$
Month 4	1 month	4000\$	18200\$	100%	4000\$	18200\$
Month 5	1 month	3600\$	21800\$	75% ✓	3800\$	22000\$
Month 6	1 month	3200\$	25000\$	0%	Not yet	Not yet

If the team is able to complete 75 percent at the end of the fifth month, then calculate the revised budget Estimate at Completion (EAC).



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Operating System

Batch: 24th

Date: 07/12/2025

Mid-Term Examination

Semester: Fall 2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions of the following.

$5 \times 6 = 30$

1. (a) What is Operating System (OS)? Write the Goals and Functions of OS. CLO1 3
(b) Explain three different types of operating system in details. CLO1 3
2. (a) Three resources R_1, R_2, R_3 each have two instances. The system state: CLO4 3
 P_1 holds 1 instance of R_1 and requests 1 instance of R_2 . P_2 holds 1 instance of R_2 and requests 1 instance of R_3 . P_3 holds 1 instance of R_3 and requests 1 instance of R_1 .

Construct the Resource Allocation Graph (RAG) and determine why cycles in RAG with multiple instances do NOT always mean deadlock.

- (b) Explain the deadlock Handling methods with proper examples. CLO5 3
3. (a) Write the criteria of CPU Scheduling. CLO5 2
(b) Processes are given with arrival and burst time on the following table. CLO5 4
The time quantum of the system is 4 units. Apply Preemptive Round Robin Scheduling Algorithm and answer the following questions.

Process	Arrival Time	Burst Time
P1	0	8
P2	0	5
P3	2	18
P4	3	11
P5	4	6
P6	5	9

- i. Draw the Gantt chart with Ready Queue.
- ii. Calculate average Waiting time.
- iii. Calculate average Turn Around time.
- iv. Calculate Throughput in percentage.

4. (a) What is process? Explain Process State Transition with figures. CLO4 3
 (b) What is PCB? Explain different kinds of light weight process with diagram? CLO4 3
5. (a) Explain the dual mode operation with figure. CLO1 2
 (b) A system has 4 processes (P0–P3) and 3 resource types (A, B, C). The current allocation, maximum requirement, and available resources are given as follows: CLO5 4

Allocation Matrix Maximum Matrix Available Resources

Proc ess	A	B	C
P0	1	0	1
P1	2	1	2
P2	3	1	1
P3	0	2	1

Proce ss	A	B	C
P0	2	1	2
P1	3	2	2
P2	3	2	2
P3	2	2	2

A	B	C
2	1	1

- (i) Calculate the NEED matrix. (Need = Max – Allocation)
 (ii) Using the Banker's Algorithm, determine whether the system is currently in a SAFE state.
 (iii) Suppose Process P1 sends a new request: Request1 = (1, 0, 1). Check whether the OS should grant or deny this request using the Banker's safety check. Provide calculations and reasoning.
6. (a) When the CPU moves from one process to another, the OS must store several important values and load new ones. Draw a diagram showing this saving/loading cycle and explain how the OS ensures the new process resumes correctly. CLO4 3
 (b) From the concept of fork() system call, determine how many times the printf functions will be executed in the following program. Explain your answer using a tree hierarchy. CLO4 3

```
#include <stdio.h>
```

```
int main(){
```

```
    fork();
```

```
    fork();
```

```
    printf("OS is easy when you understand");
```

```
    fork();
```

```
    printf("OS is hard when you don't pay attention");
```

```
    return 0;
```

```
}
```



7. In the following table, 7 processes arrive at different times with different burst times. The priorities follow the rule: **lower number = higher priority**.

CLO5 6

Process ID	Priority	Arrival Time	Burst Time
P1	3	0	6
P2	1	2	3
P3	5	1	7
P4	4	3	2
P5	2	5	5
P6	6	4	4
P7	8	6	9

- Calculate Waiting Time of all processes using FCFS Scheduling.
- Calculate Waiting Time of all processes using Priority Scheduling (Non-Preemptive).
- Compare the average waiting times and determine which algorithm completes tasks faster for this dataset.
- Explain why the chosen algorithm performs better for this specific process pattern



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 323

Final Examination

Date: 10/03/2026

Total Time: 02:00 Hours

Course Title: Numerical Analysis

Semester: Fall 2025

Total Marks: 40

Answer any Four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) Using Cramer's Rule to solve for the unknowns in three linear equations: $\begin{aligned} 5x - 2y + 3z &= 16 \\ 2x + 3y - 5z &= 2 \\ 4x - 5y + 6z &= 7 \end{aligned}$	CLO3	[5]
	(b) The matrix models a mechanical vibration system $A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$	CLO3	[5]
	(i). Find eigenvalues and corresponding eigenvectors. (ii). Briefly explain the physical significance of eigenvalues in the context of vibration or stability analysis.		
Q2.	(a) Consider that we have the following initial value problem $y' = -xy, \quad y(0) = 0.5, \text{ where } x \in [0, 0.2],$ Use the Euler method to find the approximate values of $y(0.1)$, $y(0.15)$, $y(0.2)$ and compare these values with the exact solution (find the absolute errors)	CLO3	[5]
	(b) Using the Taylor series method, find the value of y at $x = 0.1$ if $\frac{dy}{dx} = x^2y - 1, \quad y(0) = 1.$	CLO3	[5]
Q3.	Consider that we have the following initial value problem: $y' = x + y, \quad y(0) = 1$ Use the <u>Runga-Kutta</u> method of order 2 and order 4 to compute the approximate values of $y(0.25)$, $y(0.5)$ and find the absolute errors.	CLO3	[10]
Q4.	A manufacturing company is developing an automated three-axis CNC positioning system. The final position of the cutting tool depends on solving the following system of linear equations:	CLO3	[10]



HAMMARD UNIVERSITY BANGLADESH

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

Use the Gauss-Jordan Elimination Method to determine the values of x , y , and z so that the tool reaches the desired coordinate accurately. Discuss whether this method is computationally efficient for large systems.

Q5.

Consider,

CLO3 [10]

$$A = \begin{bmatrix} 1 & 6 & 2 \\ 2 & 12 & 5 \\ -1 & -3 & -1 \end{bmatrix}$$

- Show that A does not have an LU decomposition.
- Re-order the rows of A and find an LU decomposition of the new matrix.

Q6.

Consider the following data base

CLO3 [10]

X	4	6	8	10
f(x)	1	3	8	20

Find the approximate value for each of the following

- $f(4.5)$ (Forward)
- $f(9)$ (Backward)
- $f(6.4)$ (Centre)



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE325: Computer Peripherals and Interfacing

Batch: 24th

Date: 6-3-2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

Answer any five questions of the followings.

5 × 8 = 40

1. Read the following scenario and answer the questions:
A system employs an Intel 8237 DMA controller configured to perform a DMA write transfer from an I/O device to main memory. The initial memory address is 4000H, and a total of 1024 bytes are to be transferred.
 - a) Describe all the control signals associated with DMA (**MEMR, MEMW, IOR, and IOW**). Identify which control signals among them are asserted during the above operation. Justify your answer with proper explanation. CLO5 4
 - b) What is the purpose of Current Word Count Register (CWC)? Calculate the hexadecimal value that must be programmed into the CWC for the stated scenario. CLO5 4
2.
 - a) Draw a block diagram illustrating the DMA data transfer mechanism between an I/O device and memory. CLO5 4
 - b) Describe the sequence of operations involved in a DMA transfer. CLO5 4
3.
 - a) What are the difference between synchronous and asynchronous data transmission? CLO1 2
 - b) Explain I/O interface with a proper diagram. CLO1 3
 - c) Differentiate between Memory-Mapped I/O and I/O-Mapped I/O. CLO1 3
4.
 - a) List and briefly explain the various sources of interrupts in a microprocessor-based system. CLO3 2
 - b) Explain the following software interrupts in the 8086 system: CLO3 3
 - Type-0
 - Type-1
 - Type-3
 - Type-4
 - c) Define an interrupt. Describe the sequence of events that occur in the 8086 microprocessor when responding to an interrupt request. CLO3 2

- | | | |
|--|------|---|
| 5. a) Explain ISA-8 bus interface. | CLO2 | 4 |
| b) Mention the differences between PCI and PCIe. | CLO2 | 4 |
| 6. Define DMA controller. Draw and explain the pin diagram of 8237 DMA controller. | CLO5 | 8 |
| 7. The command register of the 8237 DMA controller is programmed with the binary value 0110 0101. Decode this configuration and determine the operational mode of the DMA controller. Provide justification for your answer. | CLO5 | 8 |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE323: System Analysis & Design

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

Batch: 24th
Date: 03-03-2026

Answer any five questions of the followings.

- | | | $5 \times 8 = 40$ | |
|--|------|-------------------|--|
| | CLOs | Marks | |
| 1. (a) What are the reasons of sampling in data collection? Mention the advantages of sampling. | CLO2 | 4 | |
| (b) List eight factors that must be considered when choosing output technology. | CLO3 | 4 | |
| 2. (a) Explain the five steps in interview preparation. | CLO2 | 4 | |
| (b) Illustrate diamond structure for interviewing. | CLO2 | 4 | |
| 3. (a) Compare between a logical data flow diagram and a physical data flow diagram. | CLO3 | 4 | |
| (b) Draw an entity-relationship diagram for patient treatment. | CLO3 | 4 | |
| 4. (a) When are DFDs required? What are the advantages of using a data flow approach over narrative explanations of data movement? | CLO3 | 4 | |
| (b) Create an activity diagram for room allocation in Hostel Management System. | CLO3 | 4 | |
| 5. (a) What is a context-level data flow diagram? Contrast it to a level 0 DFD. | CLO3 | 4 | |
| (b) Write the differences between traditional databases and data warehouses. | CLO3 | 4 | |
| 6. A public library wants to automate its book borrowing and return process. Currently, records are kept manually, which causes errors and delays. The new system will allow members to search books, issue books, and receive overdue notifications. The system analyst must collect requirements and create models. | CLO4 | | |
| (a) Which two fact-finding methods can be used to gather requirements? | | 2 | |
| (b) Why is requirements discovery important in analyzing the system? | | 2 | |
| (c) Identify two actors in the Use Case diagram for the system. | | 2 | |
| (d) Mention two possible use cases for this system. | | 2 | |
| 7. A company named QuickBite wants to develop an online food delivery system. Customers can browse restaurants, place orders, and make online payments. Restaurant managers can update menus, and delivery staff can update order status. The company requires proper system architecture, database design, input/output forms, and UML modeling. | CLO4 | | |
| (a) What type of system architecture is suitable for this system? | | 2 | |
| (b) Identify two main entities with attributes in the database design for the system. | | 2 | |
| (c) Mention two important input design considerations for placing orders. | | 2 | |
| (d) Which UML diagram is best for representing the system structure? | | 2 | |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 321: Operating System

Batch: 24th

Semester: Fall 2025

Date: 01/03/2026

Final Examination

Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the following.

5 × 8 = 40

1. (a) * A hard disk with 4 platters that has 2048 tracks. For each platter 1024 sectors and for each track 512-byte sector. Now calculate the total capacity of the disk. CLO1 4
- (b) Explain the following terms related to Mass Storage System: CLO1 4
- i. Magnetic Disk
 - ii. Magnetic Tape
 - iii. Solid State Disk
 - iv. Disk Structure
2. (a) Define the following terms related to disk performance: CLO1 4
- i. Seek time
 - ii. Rotational latency
 - iii. Transfer rate
 - iv. Average access time.
- (b) * Explain Moving-head Disk Mechanism with figure CLO1 4
3. (a) What is Shell Script? Why do we need Shell Script? CLO5 3
- ✓(b) Define how the followings are defined in shell script. CLO5 5
- i. If-Else
 - ii. Case
 - iii. For Loop
 - iv. Shell Arithmetic
 - v. Variables
4. (a) What is a safe state in the context of deadlock avoidance? CLO4 2
- (b) When a process requests a resource, what condition must the system check before granting the request in the resource allocation graph method of deadlock avoidance? CLO4 3
- (c) Distinguish between a safe state, an unsafe state, and a deadlock state with the help of a diagram. CLO4 3

- | | | | | |
|----|-----|--|------|---|
| 5. | (a) | Write the differences between Paging and Segmentation | CLO2 | 2 |
| | (b) | Draw the Hardware Architecture of Segmentation. Explain how the physical address is generated from Segmentation. Assume your own data to explain. | CLO2 | 6 |
| 6. | (a) | What are Contiguous and Non-Contiguous Memory Allocation? | CLO4 | 2 |
| | (b) | Define the followings with diagram (if necessary): | CLO4 | 6 |
| | | i. Internal & External Fragmentation | | |
| | | ii. Limit & Relocation Register | | |
| | | iii. Difference between Segmentation & Fragmentation. | | |
| 7. | (a) | Explain Hardware Architecture of Paging with figure. | CLO2 | 3 |
| | (b) | Solve the followings mathematical problems: | CLO2 | 5 |
| | | i. If Logical Address is 24bits, find out the Logical Address Space. | | |
| | | ii. A process size is 8MB, Physical Address Space is 16MB. If the Page Size is 4KB then calculate Total no of Page and Total no of Frames from the given information. | | |
| | | iii. Virtual Address Space is 4GB and Physical Address is 32bits. If the Page Size is 23 bits then calculate Total no of Page and Total no of Frames from the given information. | | |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
Program: B.Sc. in Computer Science and Engineering
MGT 329: Industrial and Operational Management

Final Examination

Fall 2025

6th Semester

Time: 2 Hours

24th Batch

Date: 08.03.2026

Total Marks: 40

Instructions:

- Answer any **FOUR SETS** of questions chronologically.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.

No.	Questions/Statement		Marks														
1	a)	What key elements make a forecast reliable and effective in operations management? Discuss.	4														
	b)	Compare qualitative and quantitative forecasting approaches. In what situations is each approach more suitable? Discuss.	6														
2	a)	The following table shows the activities of a project along with their durations (in weeks):<table><tr><th>Activity</th><th>Duration (weeks)</th></tr><tr><td>1-2</td><td>9</td></tr><tr><td>1-3</td><td>6</td></tr><tr><td>2-4</td><td>5</td></tr><tr><td>3-5</td><td>7</td></tr><tr><td>4-6</td><td>4 <i>extra</i></td></tr><tr><td>5-6</td><td>10</td></tr></table> Requirements: 1. Draw the network diagram of the project.	Activity	Duration (weeks)	1-2	9	1-3	6	2-4	5	3-5	7	4-6	4 <i>extra</i>	5-6	10	10
Activity	Duration (weeks)																
1-2	9																
1-3	6																
2-4	5																
3-5	7																
4-6	4 <i>extra</i>																
5-6	10																

		2. Determine the earliest start time (EST), earliest finish time (EFT), late start time (LST), and late finish time (LFT). 3. Identify the critical path and calculate the total project duration. 4. Calculate the total float of each activity.																			
3	a)	Differentiate between CPM (Critical Path Method) and PERT (Program Evaluation and Review Technique).	3																		
	b)	A company has recorded its actual sales figures alongside forecasted sales for four consecutive quarters of the year, as presented in the table below: - - - - <table><tr><th>Quarter</th><th>Actual Sales (Units)</th><th>Forecasted Sales (Units)</th></tr><tr><td>Q1</td><td>420</td><td>400</td></tr><tr><td>Q2</td><td>450</td><td>440</td></tr><tr><td>Q3</td><td>470</td><td>460</td></tr><tr><td>Q4</td><td>500</td><td>490</td></tr></table> Requirements: Based on the above data, perform the following tasks: 1. Calculate the Mean Absolute Deviation (MAD). 2. Calculate the Mean Squared Error (MSE). 3. Calculate the Mean Absolute Percentage Error (MAPE). 4. Briefly comment on the accuracy of the forecast.	Quarter	Actual Sales (Units)	Forecasted Sales (Units)	Q1	420	400	Q2	450	440	Q3	470	460	Q4	500	490	7			
Quarter	Actual Sales (Units)	Forecasted Sales (Units)																			
Q1	420	400																			
Q2	450	440																			
Q3	470	460																			
Q4	500	490																			
4		Given the following sales data of Unilever Bangladesh Limited: <table><tr><th>Week</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><th>Sales</th><td>39</td><td>44</td><td>40</td><td>45</td><td>38</td><td>43</td><td>39</td><td>?</td></tr></table> (a) Forecast for <u>Week 8</u> using the Simple Moving Average (SMA) method and discuss why it is not trustworthy all the time. (b) Forecast sales using the 4-week Weighted Moving Average (WMA) method with Weights 0.4, 0.3, 0.2, 0.1.	Week	1	2	3	4	5	6	7	8	Sales	39	44	40	45	38	43	39	?	10
Week	1	2	3	4	5	6	7	8													
Sales	39	44	40	45	38	43	39	?													

5	a)	Why do organizations engage in product and service design or redesign? Identify and explain at least five major reasons with suitable examples.	4
	b)	What key issues should managers consider in product and service design? Discuss them with a practical example.	6